Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Two equal 10 Ω resistors are connected in parallel. Find equivalent resistance, then compare with a series connection of the same pair.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: P = IV; E = PΔt; P = I^2R = V^2/R.
Constant power for E = PΔt; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power VI=12 W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Marzano check - circle: ① ② ③

03 Represent and interpret

Two motors use the same energy but one takes half the time. Compare average power.

Marzano check - circle: ① ② ③

04 Solve and justify

A 12 V device draws .75 A for 4 minutes. Find power and energy in joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Marzano check - circle: 1 2 3

06 Choose a method

A shallow-water model uses $v=\sqrt{(gh)}$. With $g=10$, $h=40$ m and travel distance 6 km, find speed and travel time. State a model limitation.

Marzano check - circle: 1 2 3

07 Exit check

A 6 Ω resistor carries 2 A. Find its power using two equivalent equations.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 12 V motor draws 1.5 A for 40 s and delivers 432 J of useful mechanical energy. Find electrical input energy, average useful power and efficiency.

Marzano check - circle: 1 2 3